Lab: Designing Your Testing Agent Protocol

Objective

Design a multi-agent testing strategy for your invention project that deliberately incorporates diverse perspectives, manages creator bias, and includes adversarial testing. By the end of this lab, you will have a testing protocol that specifies who tests, what they look for, and how their observations are collected and interpreted.

Prerequisites

- Familiarity with the creator-evaluator transition, distributed sensing, and adversarial testing from this module's lecture
- Your invention project and prototype specification from Module 01
- At least one other person available for a brief testing exercise (or the ability to simulate external perspectives)

Materials

- Your prototype (physical, digital, or conceptual)
- Notebook or digital document for recording observations
- Timer

Part 1: Self-Assessment — Your Agent Configuration (10 minutes)

Goal: Map your own generative model as a testing agent, identifying your priors, attention patterns, and potential blind spots.

Answer honestly — the value of this exercise depends on self-awareness:

Your priors about your invention:

What do you believe most strongly about your invention? What are you most confident will work?

Write your response here

What are you least sure about? Where is your uncertainty highest?

Write your response here

Your attention profile:

When you look at your prototype, what do you naturally focus on first? (Mechanism? Aesthetics? User interface? Materials? Cost?)

Write your response here

What aspects of the prototype do you tend to overlook or check last?

Write your response here

Your bias risk assessment:

Bias	Risk Level (Low/Med/High)	Why?
Confirmation bias (interpreting ambiguous results as positive)		

Write your response here

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Write your response here

| | Sunk cost bias (reluctance to abandon failed approaches) |

Write your response here

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Write your response here

| | Anchoring bias (over-weighting early test results) |

Write your response here

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Write your response here

| | Expertise blindness (assuming users share your knowledge) |

Write your response here

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Write your response here

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Part 2: Design Your Testing Agent Team (15 minutes)

Goal: Identify the different types of testing agents your invention needs and what each brings to the evaluation.

For each agent type, specify who they are, what generative model they bring, and what prediction errors they are likely to generate:

Agent Type 1: The Naive User - Who: Someone with no knowledge of your invention's domain or mechanism - What they test: Basic usability, first impressions, intuitive understanding - Who specifically could you recruit for this role?

Write your response here

- What instructions would you give them?

Write your response here

Agent Type 2: The Domain Expert - Who: Someone with deep expertise in the problem your invention solves - What they test: Technical viability, competitive comparison, unmet needs - Who specifically could you recruit for this role?

Write your response here

- What questions would you ask them?

Write your response here

Agent Type 3: The Adversarial Tester - Who: Someone given explicit permission and incentive to break the prototype - What they test: Failure modes, edge cases, worst-case scenarios - Who specifically could you recruit for this role?

Write your response here

- What constraints would you set (to prevent actual danger while allowing stress testing)?

Write your response here

Agent Type 4: The Target Customer - Who: Someone who matches the profile of your intended user - What they test: Value proposition, willingness to use/buy, integration with existing habits - Who specifically could you recruit for this role?

Write your response here

- What scenario would you set up for their test?

Write your response here

Part 3: Run a Mini Test Session (20 minutes)

Goal: Conduct a brief test of your prototype with at least one external agent, practicing the skills of observation without intervention.

If you have access to another person, ask them to interact with your prototype (or a description/sketch of it) for 5-10 minutes. If working alone, simulate an external agent by deliberately adopting a different perspective (e.g., imagine you are a complete novice encountering this for the first time).

Rules for the test: 1. Give the tester the prototype with minimal explanation 2. Do NOT explain how it works unless they are stuck and frustrated 3. Observe silently — record what they do, not what you expected them to do 4. Note every moment of confusion, surprise, or unexpected behavior

Observation Record:

Time	What the tester did	What you expected them to do	Prediction error (difference)

Write your response here

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Write your response here

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Write your response here

Write your response here

Write your response here

Write your response here

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Post-test debrief (ask the tester):

What did you think this was supposed to do?

Write your response here

What was confusing or surprising?

Write your response here

What would you change?

Write your response here

Part 4: Analyze Agent Divergence (10 minutes)

Goal: Compare your own observations of the prototype with the external agent's observations and analyze the differences.

Review the prediction errors from Part 3. For each significant discrepancy between your expectations and the tester's behavior:

Prediction Error	Was this a surprise to you?	What does it reveal about your generative model?	What should you update?

Write your response here

Write your response here

Write your response here

Write your response here

Write your response here

Write your response here

Write your response here

Write your response here

Write your response here

Write your response here

Write your response here

Write your response here

Now categorize your updates:

- **Model-confirming observations** (tester did what you expected):

Write your response here

- **Parameter-updating observations** (tester struggled with specific details, but the basic approach works):

Write your response here

- **Structure-updating observations** (tester's behavior suggests a fundamental assumption is wrong):

Write your response here

Part 5: Write Your Testing Protocol (10 minutes)

Goal: Compile your analysis into a formal testing protocol document.

Testing Protocol for:

Write your response here

Pre-registration statement (commit to these before testing): - Hypothesis being tested:

Write your response here

- Specific success criteria (quantitative if possible):

Write your response here

- Specific failure criteria:

Write your response here

- What result would make you change your design?

Write your response here

Agent deployment plan:

Phase	Agent Type	Number of Testers	What They Test	How You Collect Data
1				

Write your response here

Write your response here

Write your response here

Write your response here

| | 2 |

Write your response here

|

Write your response here

|

Write your response here

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Write your response here

| | 3 |

Write your response here

Write your response here

Write your response here

Write your response here

Bias mitigation measures:

Write your response here

Discussion and Debrief

1. **The hardest part**: What was most difficult about observing someone else interact with your invention without intervening? What does this reveal about your priors?

2. **Surprise value**: What was the single most surprising observation from your test session? Was it more valuable than the confirming observations?

3. **Agent diversity**: If you could add one more type of testing agent to your protocol, who would it be and why?

4. **Pre-registration challenge**: Was it difficult to commit to specific failure criteria? Why might inventors resist pre-registering what would count as failure?

5. **Adversarial comfort**: How comfortable are you with the idea of someone deliberately trying to break your invention? What does your comfort level reveal about your relationship to your generative model?

Write your response here